L4

CLASS X - SCIENCE

HUMAN EYE AND THE COLOURFUL WORLD

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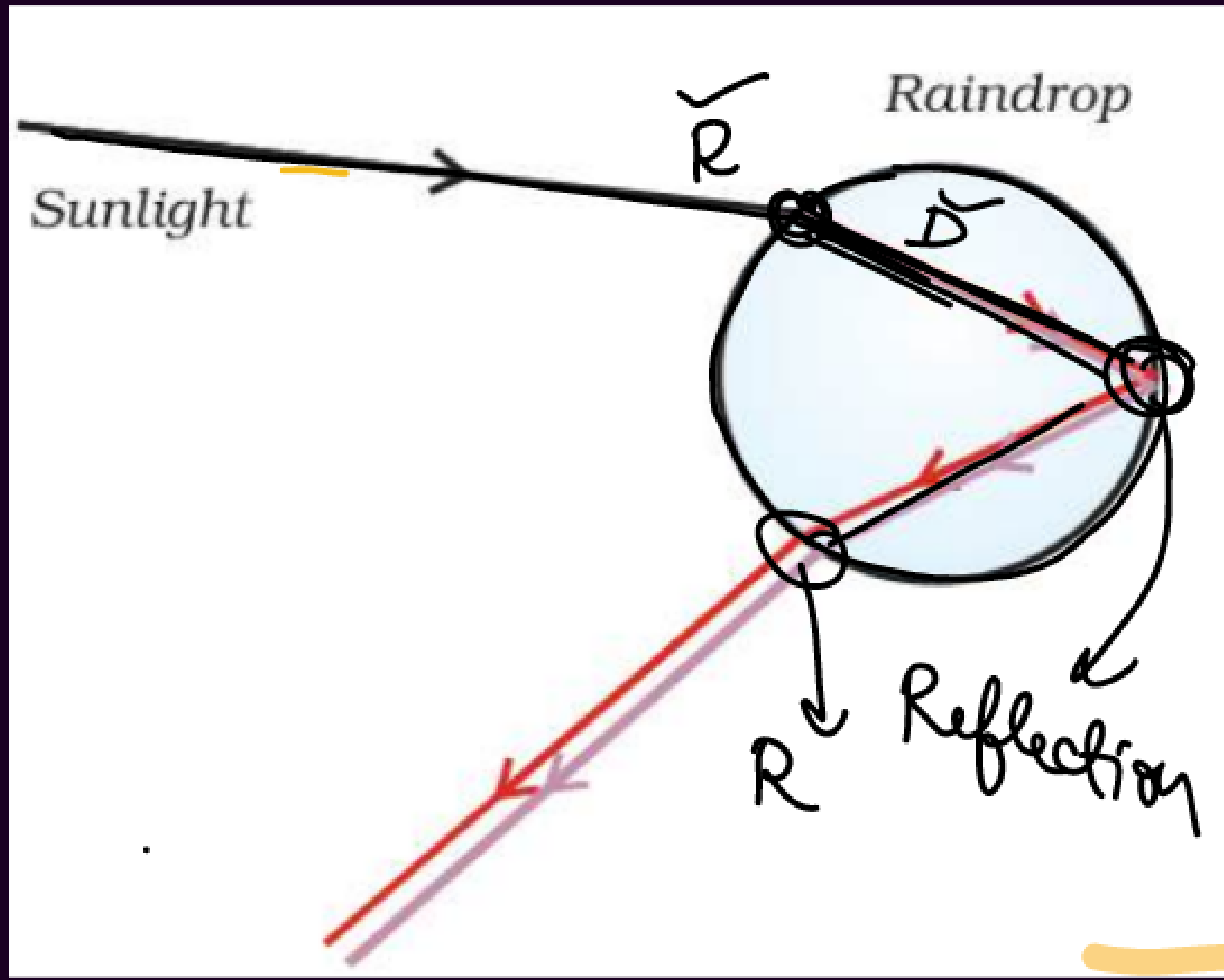
Rainbow formation

RDRR

- A rainbow is a natural spectrum appearing in the sky after a rain shower
- It is caused by dispersion of sunlight by tiny water droplets, present in the atmosphere.
- The water droplets act like small prisms. They refract and disperse the incident sunlight, then reflect it internally, and finally refract it again when it comes out of the raindrop



"A rainbow is always formed in a direction opposite to that of the Sun."

Soch ke batao

Assertion (A): The rainbow is a natural spectrum of sunlight in the sky.

Reason (R): Rainbow is formed in the sky when the sun is overhead and water droplets are present in air. [CBSE 2024]

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.

Soch ke batao

Q. Which of the following phenomena of light are involved in the formation of a rainbow?

(a) Reflection, refraction and dispersion

(b) Refraction, dispersion and total internal reflection

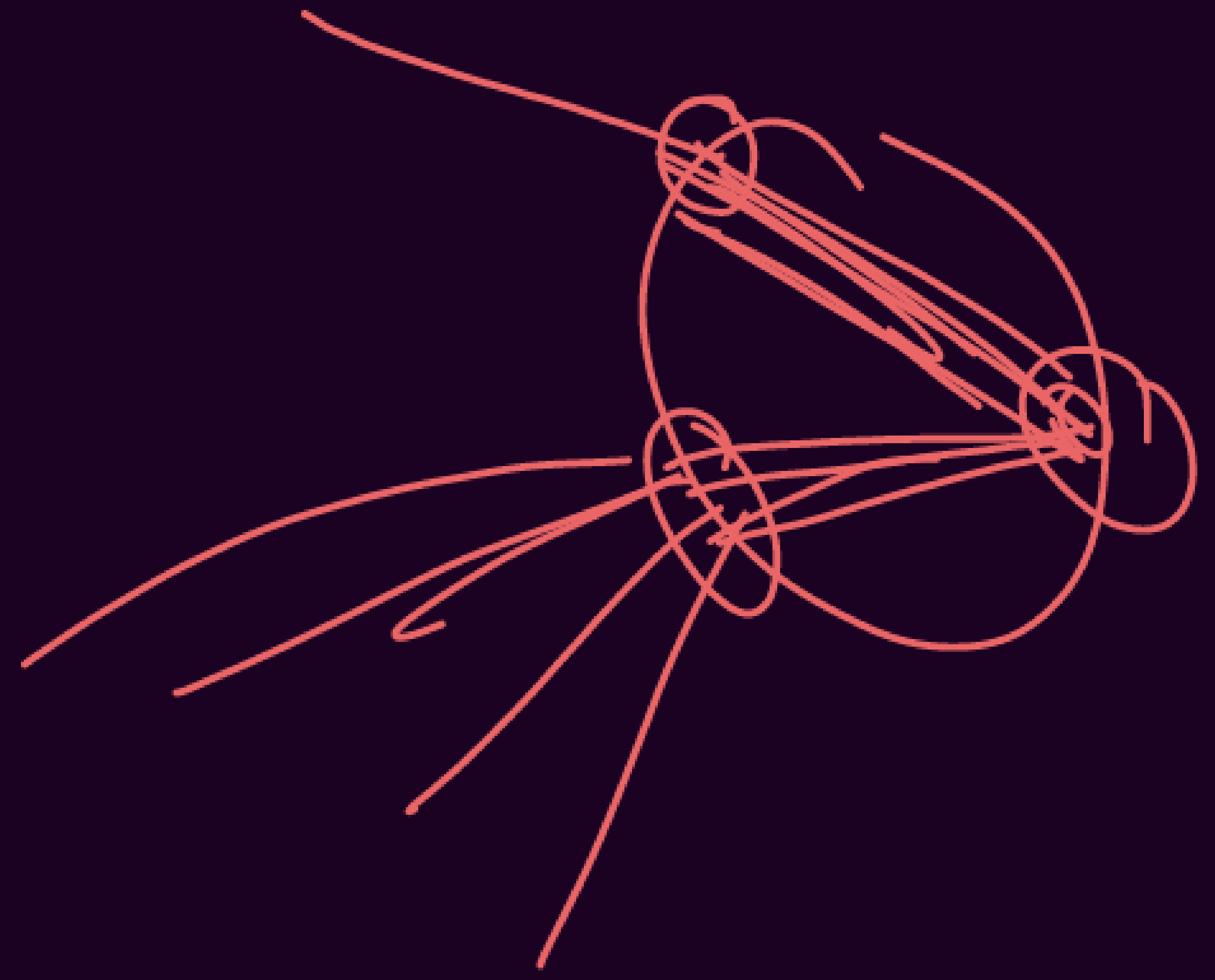
(c) Refraction, dispersion and internal reflection

(d) Dispersion, scattering and total internal reflection

Internal

Soch ke batao

Q. Draw a ray diagram to show the formation of a rainbow in the sky. On this diagram mark A - where dispersion of light occurs, B - where internal reflection of light occurs and C - where refraction of light occurs. List two necessary conditions to observe a rainbow. [CBSE 2024]



Soch ke batao

Q. In a lab, two triangular prisms made of different materials are used to study the deviation of a ray of light. The angle of incidence is kept constant at 45° for both. The following deviations were observed:

Material	Angle of Deviation
Prism A (Glass)	18°
Prism B (Plastic)	12°

A

(A) Which prism has a higher refractive index? Justify your answer.

(B) How does this property relate to the choice of material in making optical instruments like periscopes or binoculars?

(A) Prism A (Glass) has a higher refractive index.

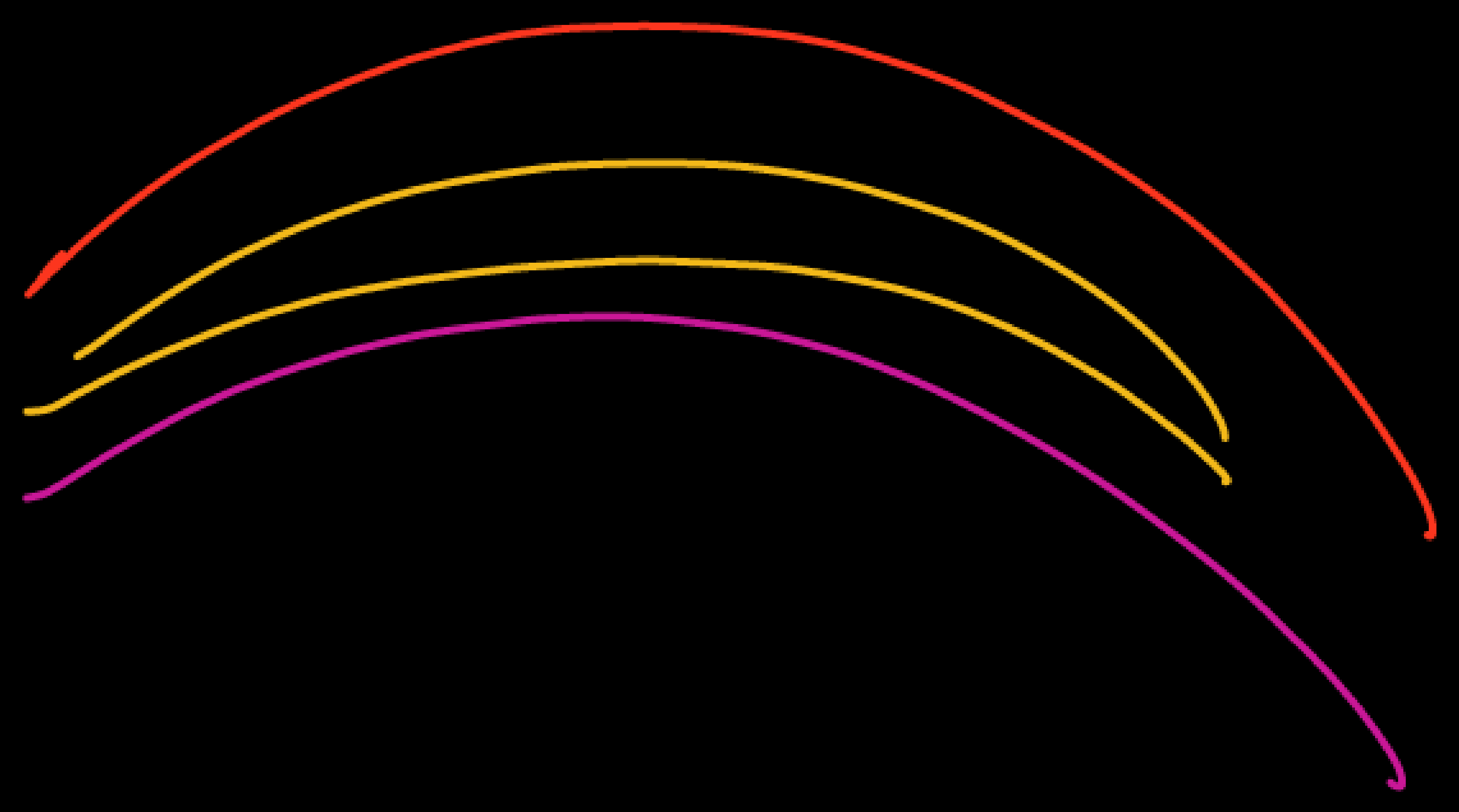
For the same angle of incidence, a prism that produces a greater angle of deviation has a higher refractive index. Since Prism A shows 18° deviation (greater than 12°), it is optically denser.

(B) Materials with higher refractive index bend light more effectively and allow better internal reflection. Hence, high refractive index materials like glass are preferred in optical instruments such as periscopes and binoculars for efficient light bending and clearer images.

Soch ke batao

Q. (a) Explain why a rainbow is always formed in a direction opposite to the Sun.
(b) Why does red colour appear at the top of the rainbow and violet at the bottom?

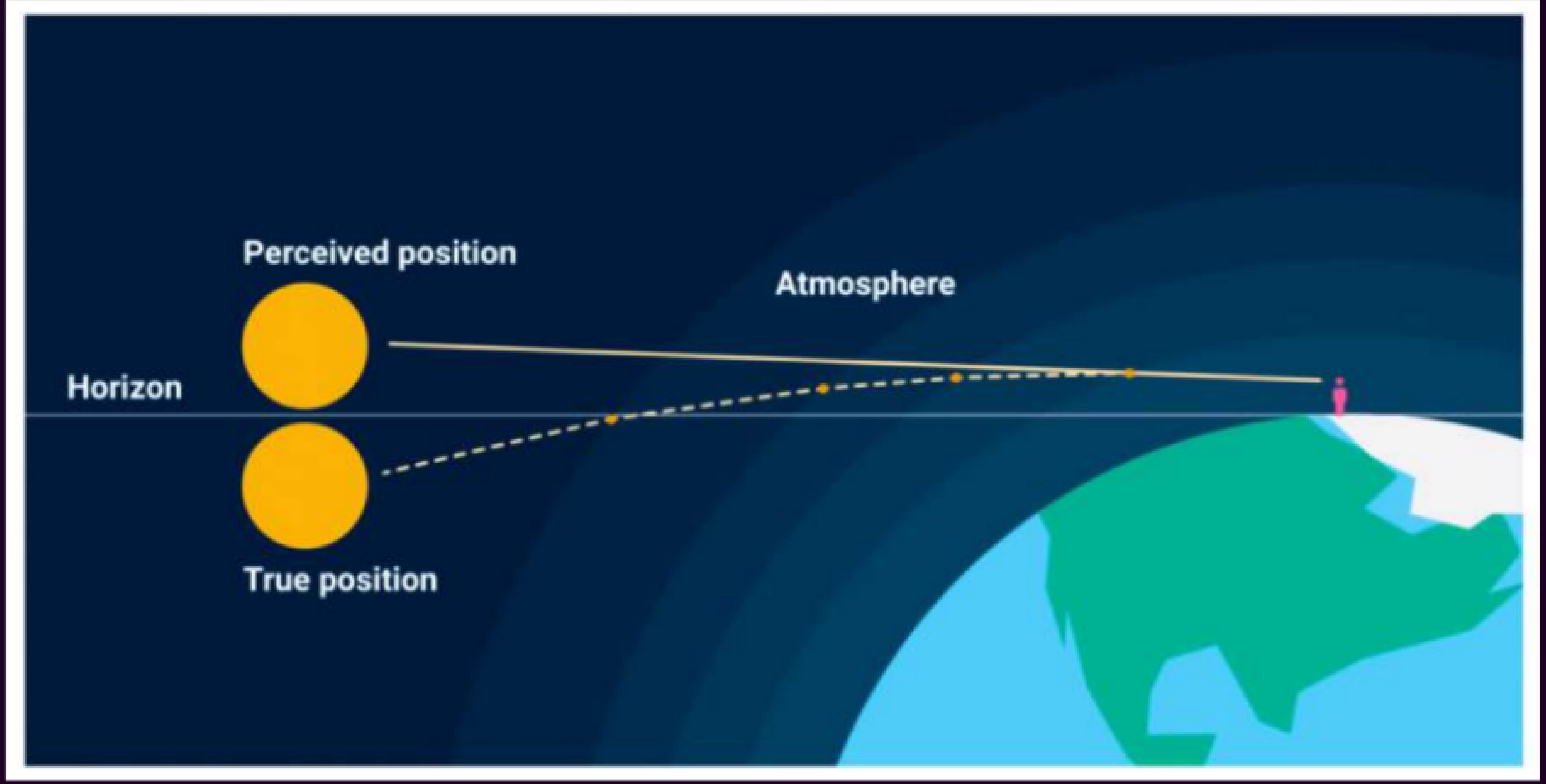
(a) A rainbow is seen when the Sun is behind the observer and rain droplets are in front. The light undergoes refraction, internal reflection, and dispersion inside the raindrops, which leads to the formation of the rainbow in the opposite direction.
(b) Because red light deviates the least and violet light deviates the most during dispersion through water droplets.

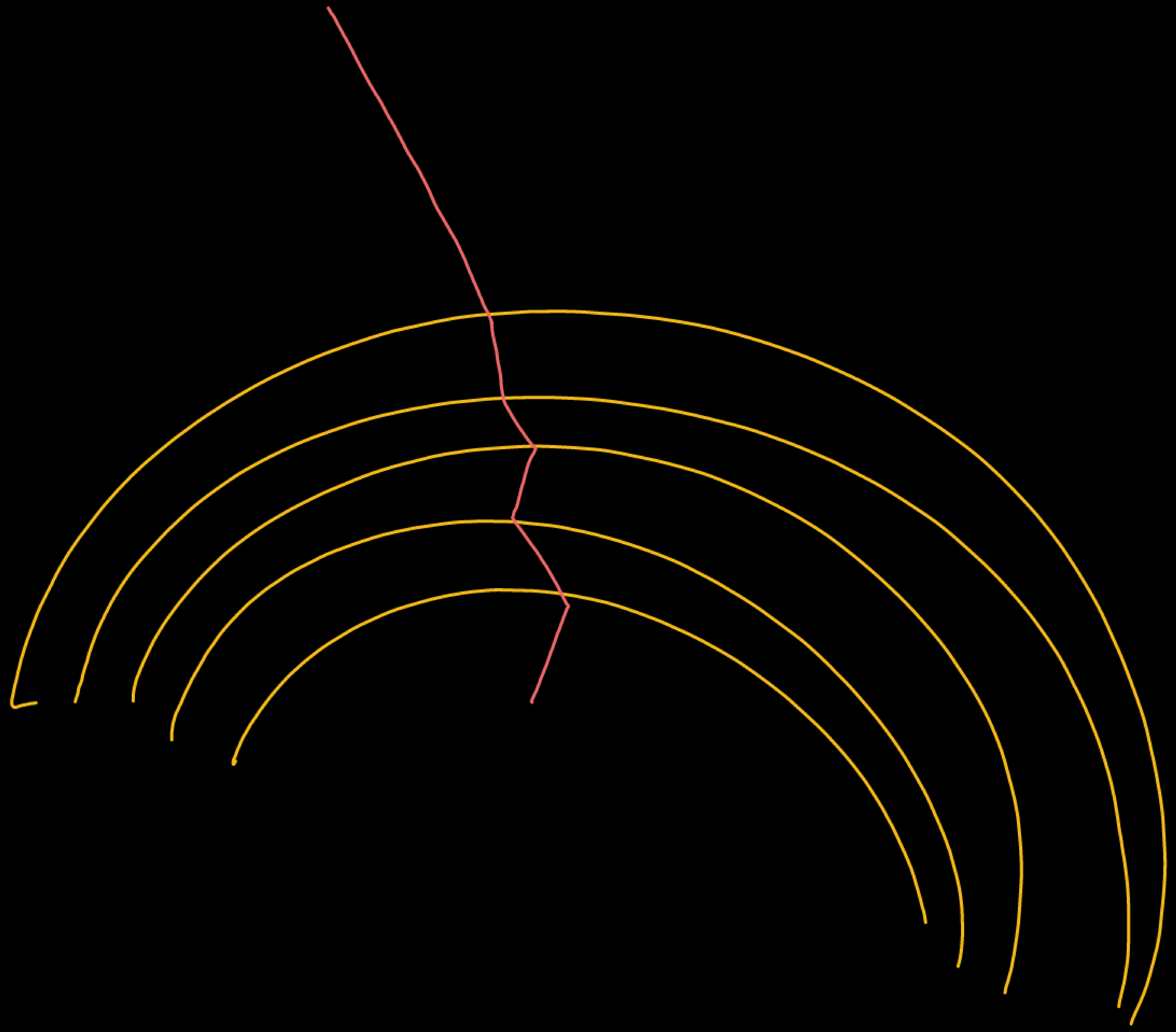


Have you ever thought? why the colour of sky is BLUE or RED?

Atmospheric Refraction

- *It is the bending of light or other electromagnetic waves as they pass through the Earth's atmosphere.*
- It occurs because the atmosphere's layers have **different optical densities**, which is due to variations in air density with height.

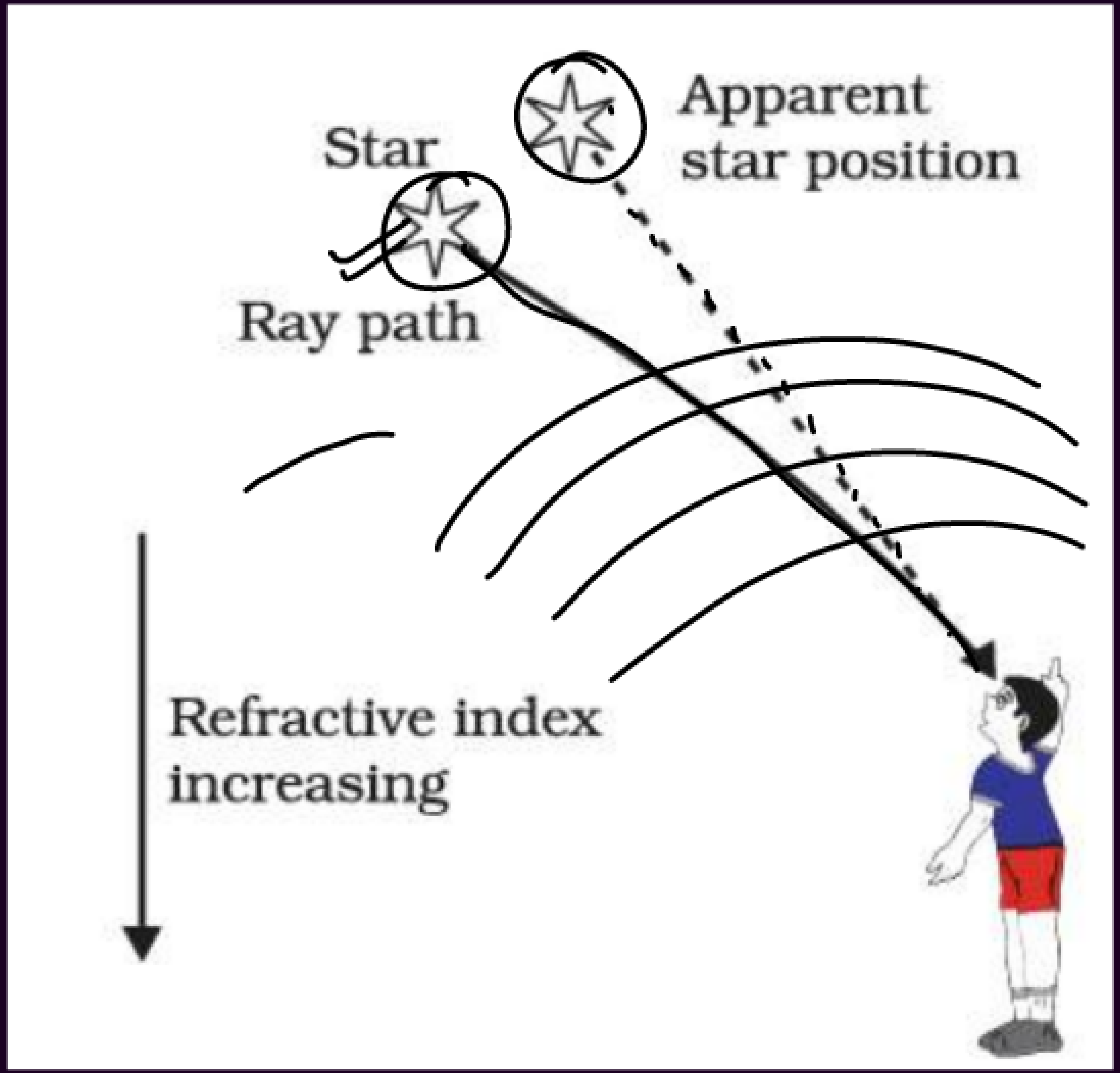


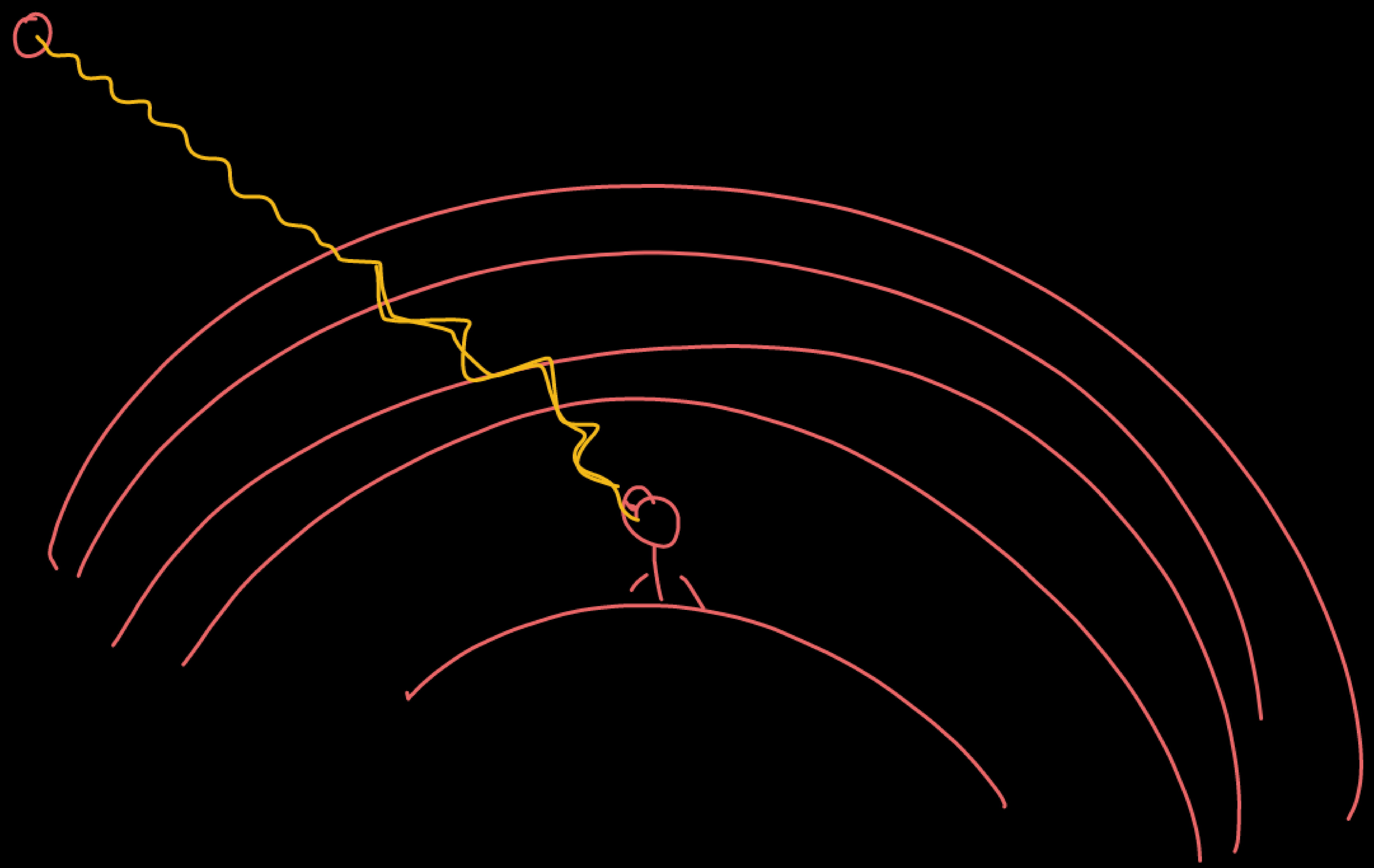


Effects of Atmospheric refraction

Twinkling of Stars

- The **twinkling of stars is caused by atmospheric refraction of starlight.**
- As starlight enters the Earth's atmosphere, it passes through layers of air having different densities and refractive indices, causing the light to bend continuously.
- Due to this refraction, the **apparent position** of the star keeps changing slightly and the star appears higher than its actual position.
- Since stars are very far away and act like point sources of light, the amount of light entering our eyes changes continuously.
- As a result, the star sometimes appears brighter and sometimes dimmer, which is observed as the twinkling of stars.



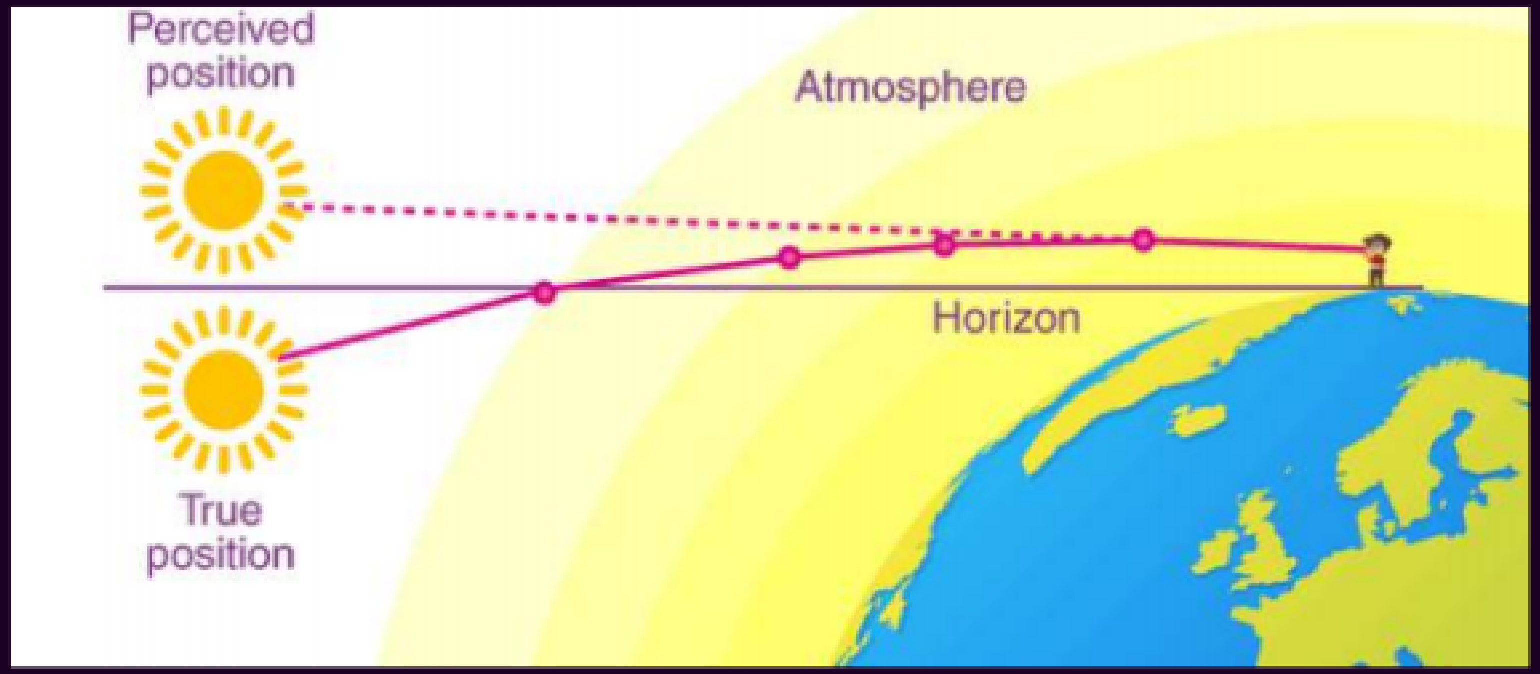


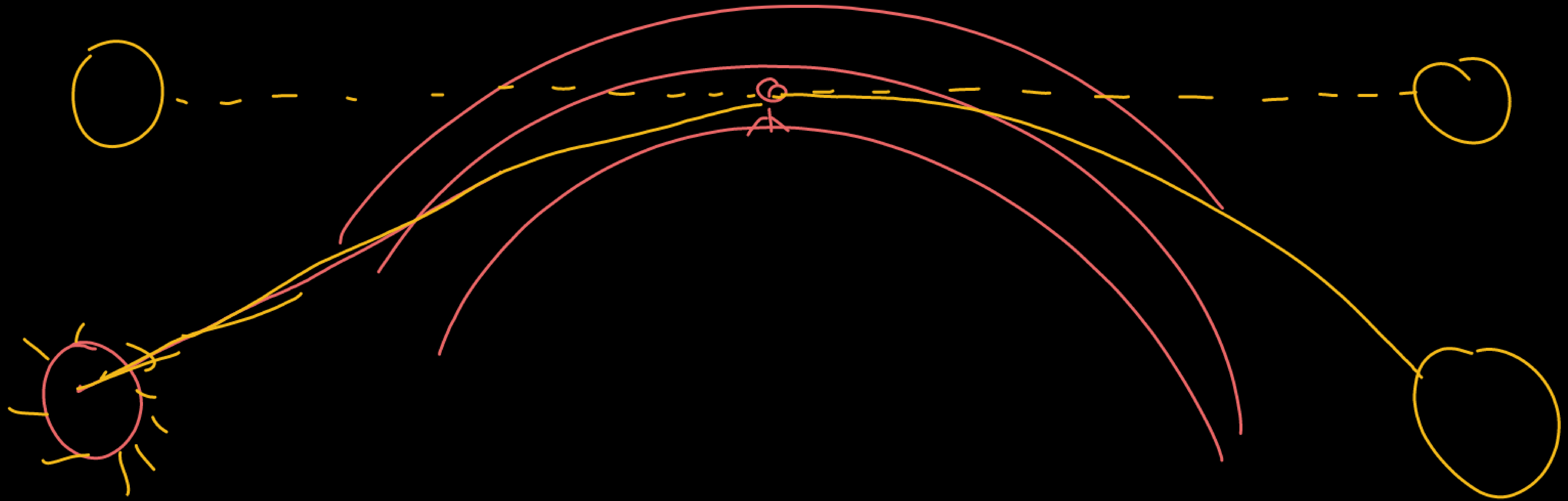
Why don’t the planets twinkle?

- Planets do not twinkle because they are much closer to the Earth and appear as extended sources of light.
- The light coming from different parts of a planet averages out the variations caused by atmospheric refraction.
- Therefore, planets shine steadily and do not show the twinkling effect.

Advance sunrise & delayed sunset

- The Sun is visible about 2 minutes before actual sunrise and about 2 minutes after actual sunset due to atmospheric refraction.
- Atmospheric refraction bends sunlight towards the Earth.
- As a result, the Sun appears slightly higher than its actual position.
- The apparent flattening of the Sun at sunrise and sunset is also due to atmospheric refraction.





Illusion due to atmospheric refraction

HW

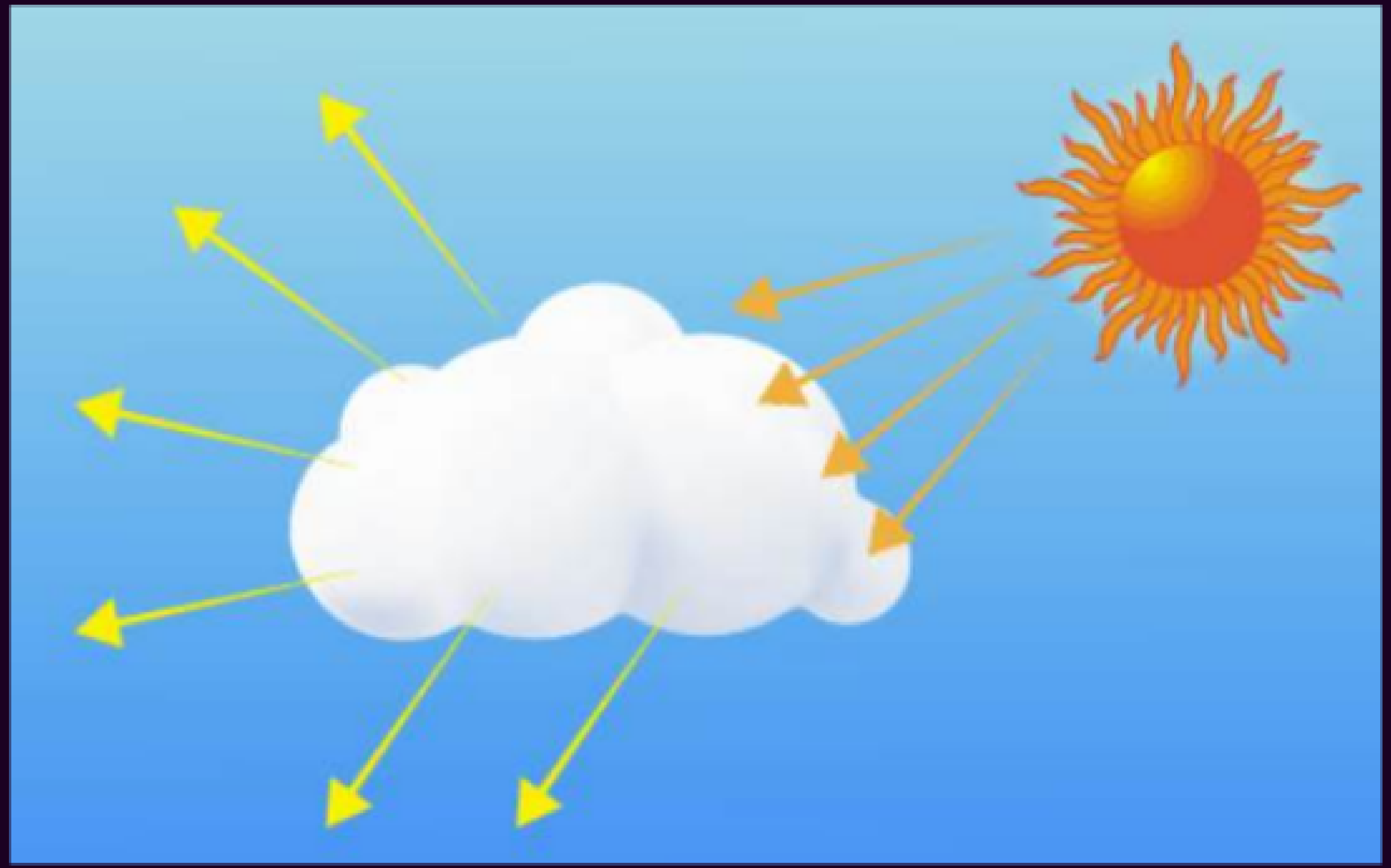
Mirage

Sundogs

Looming

Scattering of Light

- The phenomenon in which light is redirected in different directions when it strikes small particles of a medium is called scattering of light.
- The amount and colour of light scattered depend mainly on the **size of the scattering particles** and the **wavelength of light**.



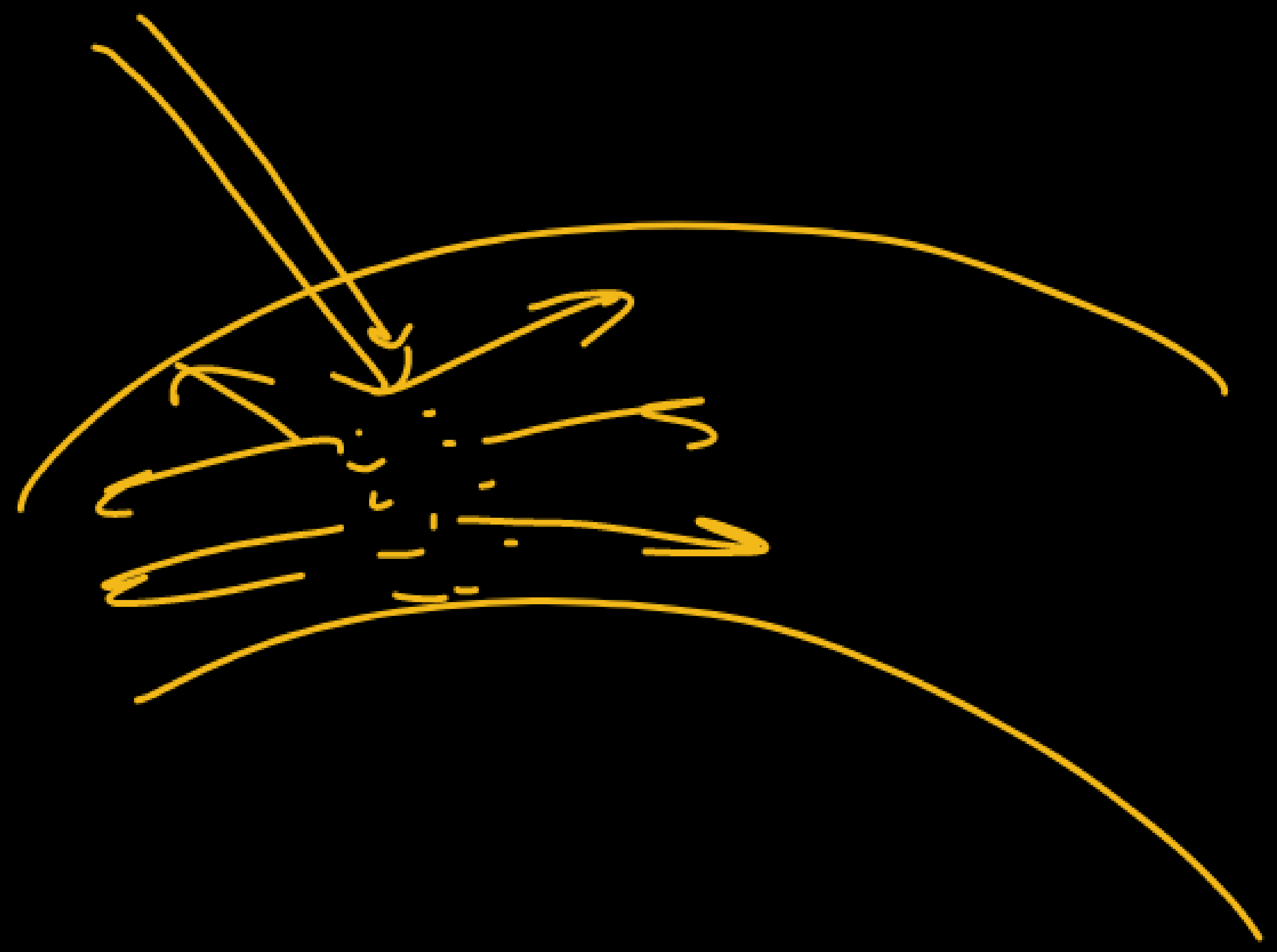
Effect of Particle Size:

- Very fine particles scatter mainly shorter wavelengths, such as blue light.
- Larger particles can scatter light of longer wavelengths as well.
- Very large particles scatter almost all colours of light, making the scattered light appear white.

Shorter the wavelength of light, greater is its scattering.

Blue sky during daylight → BLACK

- The molecules of air and fine particles in the atmosphere scatter sunlight.
- Blue light has a shorter wavelength, so it is scattered more than red light.
- The scattered blue light enters our eyes, making the sky appear blue.
- If there were no atmosphere, the sky would appear dark.



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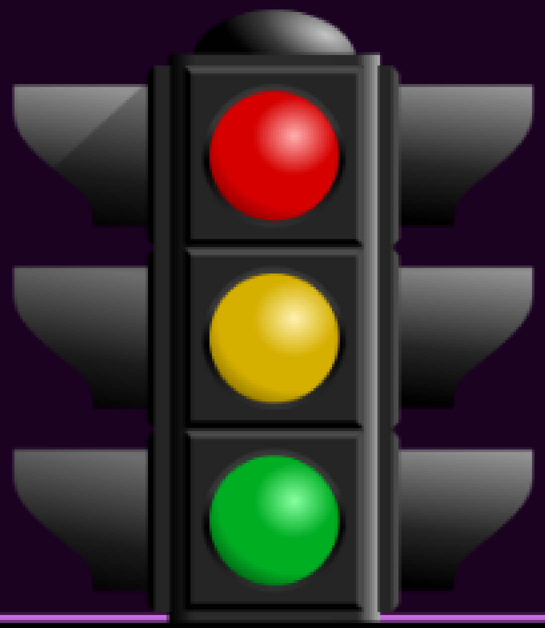
Red sky at the time of sunrise and sunset

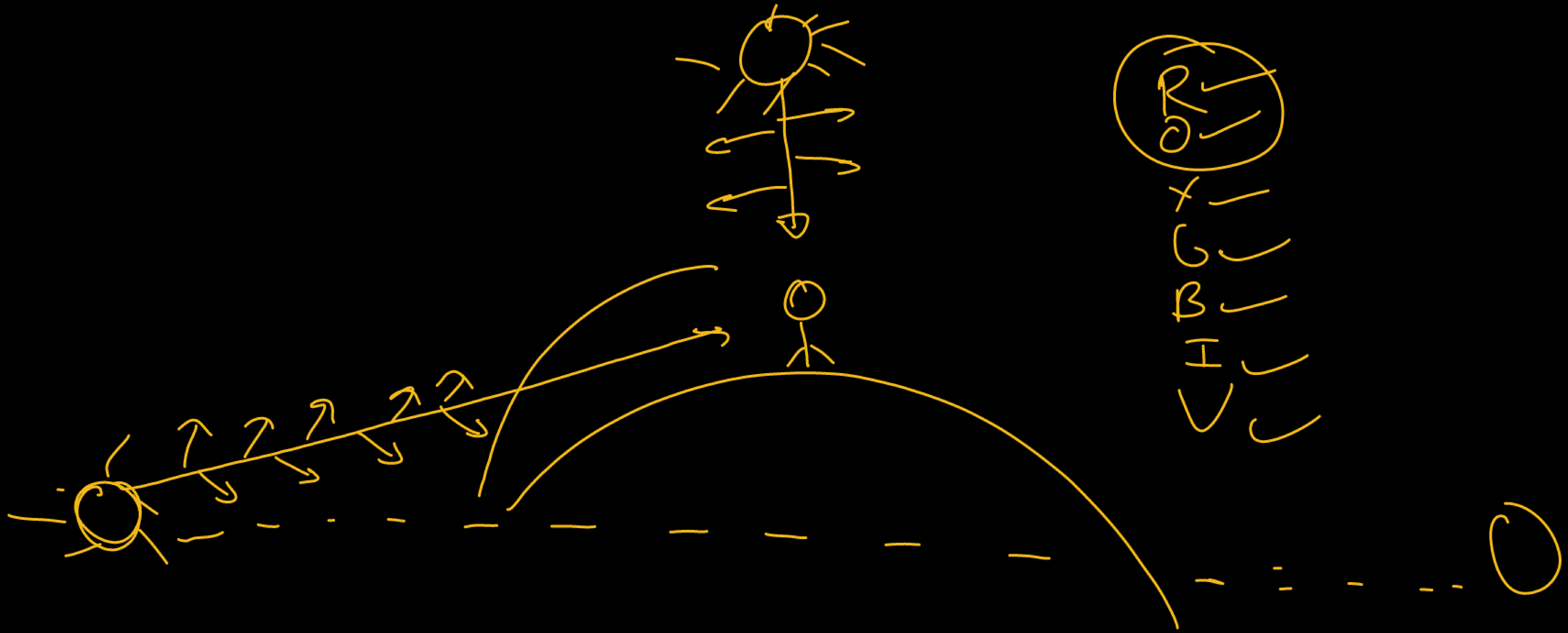
- During sunrise and sunset, sunlight travels a longer distance through the atmosphere.
- The shorter wavelength blue light gets scattered away by air particles.
- Red light, having a longer wavelength, is scattered less and reaches our eyes.
- Therefore, the Sun appears reddish during sunrise and sunset.

Q. How can we explain the reddish appearance of sun at sunrise or sunset? Why does it not appear red at noon?

Q **Why are danger signals red?**

- Red light has the longest wavelength.
- It is scattered the least by fog and smoke.
- Therefore, red colour is used for danger signals.

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Without Earth's atmosphere, no scattering of light would occur, and the sky would appear dark, similar to what passengers observe at high altitudes.

Tyndall effect

- The Earth's atmosphere contains tiny particles like dust, smoke, water droplets, and air molecules.
- When light passes through these particles, its path becomes visible due to scattering.
- **This scattering of light by colloidal particles is called the Tyndall effect.**
- It can be seen when sunlight enters a smoke-filled room or passes through a dense forest.
- Very fine particles scatter blue light more, while larger particles may scatter white light.

Soch ke batao

Q. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is: [CBSE 2024]

(a) Red
(b) Orange
(c) Blue
(d) Yellow

Soch ke batao

Q. Assertion (A): Sky appears blue in the day time.
Reason (R): White light is composed of seven colours. [CBSE 2021]

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, and R is not the correct explanation of A.
(c) (A) is true, but (R) is false.
(d) (A) is false but (R) is true.

Soch ke batao

Q. The face of the moon that is visible to us is called as the near side and the face of the moon which is invisible to us is called as far side. What colour would the sky appear to an astronaut standing on the "far side" of the Moon and why?

(a) blue, as the Moon's atmosphere scatters sunlight just like Earth
(b) white, as the Moon's surface reflect all the light that falls on it
(c) black, as there is no atmosphere on Moon to scatter sunlight
(d) black, as sunlight does not fall on the far side of the Moon

Soch ke batao

Q. Twinkling of stars is due to atmospheric

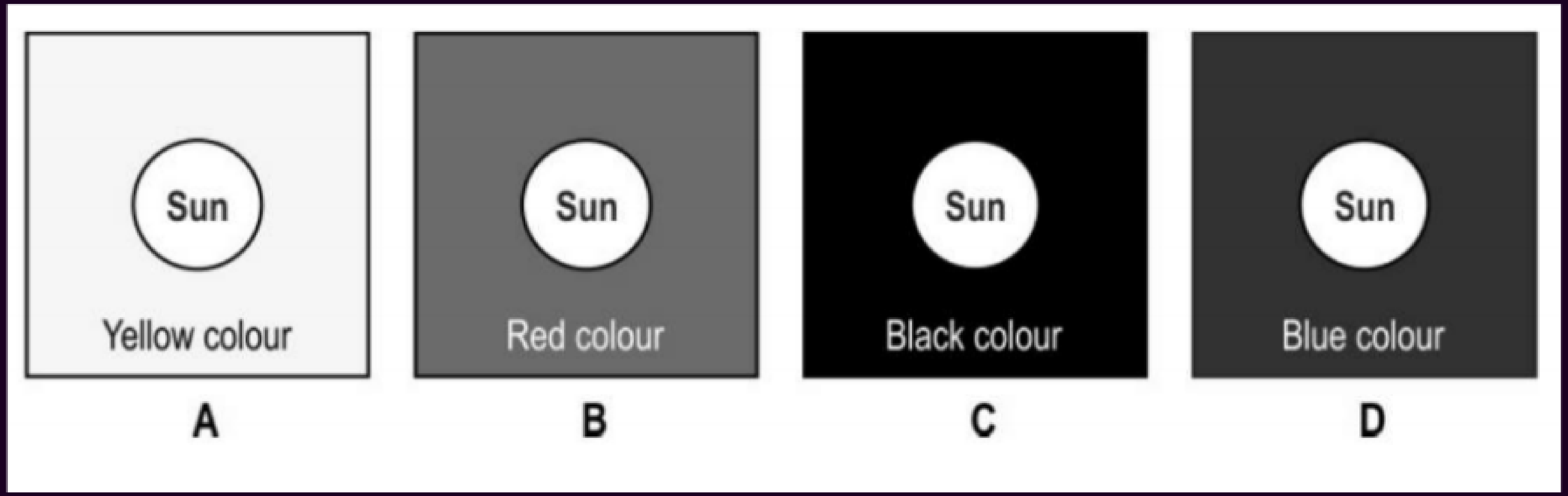
(a) dispersion of light by water droplets
(b) refraction of light by different layers of varying refractive indices
(c) scattering of light by dust particles
(d) internal reflection of light by clouds

Soch ke batao

Q. On a windy night, stars appear to twinkle more. Why?
A) More light scattering
B) Light gets absorbed
C) Greater variation in atmospheric refraction
D) Stars move faster

Soch ke batao

Q. If the Earth did not have an atmosphere, which of the following shows what you would see if you look at the Sun?



Soch ke batao

Q. Name and explain the phenomenon of light due to which the path of a beam of light becomes visible when it enters a smoke filled room through a small hole. Also state the dependence of colour of the light we receive on the size of the particle of the medium through which the beam of light passes. (CBSE 2024)

Answer:

The phenomenon responsible is the Tyndall Effect. It is the scattering of light by tiny particles present in a medium, due to which the path of light becomes visible in a smoke-filled room.

Dependence on particle size:

- Very fine particles scatter mainly blue light.
- Larger particles scatter light of longer wavelengths.
- Very large particles scatter all colours almost equally, making the scattered light appear white.

Soch ke batao

Q. Define atmospheric refraction. Explain the following effects with reason:
(i) Twinkling of stars
(ii) Advanced sunrise and delayed sunset

Atmospheric refraction: The bending of light while passing through different layers of the atmosphere due to change in refractive index.

(i) Twinkling of stars:

Stars twinkle because their light undergoes continuous atmospheric refraction through changing layers of air.

(ii) Advanced sunrise and delayed sunset:

Due to atmospheric refraction, the Sun appears higher than its actual position. Hence, it is seen about 2 minutes before sunrise and 2 minutes after sunset.

Soch ke batao

Q. Is the position of a star as seen by us its true position? Justify your answer.

No, the apparent position of a star is not its true position.
Due to atmospheric refraction, starlight bends continuously while passing through the Earth's atmosphere. As a result, the star appears slightly higher than its actual position.

Soch ke batao

Q. Give reasons for the following.
(A) Danger signals installed at airports and at the top of tall buildings are of red colour.
(B) The sky appears dark to the passengers flying at very high altitudes.
(C) The path of a beam of light passing through a colloidal solution is visible. [CBSE 2023]
Answer:

(A)
Danger signals are red because red light has the longest wavelength and is scattered the least by fog, smoke, and dust. Therefore, it can be seen clearly from a long distance.
(B)
At very high altitudes, the atmosphere is very thin and scattering of sunlight is very less. Hence, the sky appears dark to passengers flying at high altitudes.
(C)
The path of light becomes visible in a colloidal solution due to scattering of light by colloidal particles. This phenomenon is called the Tyndall effect.

Soch ke batao

Q. While watching a sunset, Arjun observed that the Sun appeared oval in shape and stayed visible for some time even after it had gone below the horizon.
1: What phenomenon is responsible for this?
2: Why does the Sun look oval?
3: How long does atmospheric refraction affect the visibility of the Sun?

Ans. 1.Atmospheric refraction is responsible for this observation.
2. The Sun looks oval because the lower part is more refracted than the upper part due to varying air densities, causing unequal bending of light.
3. Atmospheric refraction affects the visibility of the Sun for about 2 minutes after sunset.

Soch ke batao

Q. Nanda saw rays of sunlight entering into a dark room as shown below. He then did something to the air in the room after which he was NOT able to see the rays of sunlight in the room.

1. What is it that Nanda could have done to make the rays of sunlight invisible? Justify your answer.

Mars's atmosphere is composed mainly of carbon dioxide, nitrogen and argon and negligible amounts of oxygen, water vapour and methane.

Using the information given in the sentence above and knowledge about how rainbows are formed on Earth, explain why rainbow formation is impossible on Mars.

2. Space is mostly vacuum, devoid of any medium.

(a) What colour does the Sun appear to the astronauts on International Space Station?

(b) Give reason for your answer to (a).

1. He removed the dust particles from the air (for example, by filtering or cleaning the air).
Justification:
Sunlight rays are visible due to scattering by dust particles in the air. When dust particles are removed, scattering stops and the rays become invisible.
A rainbow forms due to dispersion of sunlight in water droplets present in the atmosphere.
Mars has negligible water vapour, so water droplets are absent. Therefore, rainbow formation is not possible on Mars.
2.
(a) The Sun appears white to astronauts on the International Space Station.
(b) In space, there is no atmosphere to scatter sunlight. Hence, all colours reach the eye together, making the Sun appear white.